

ML FOR MECH ENG (CoEP)

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Outline

① SESSION 14: KNN

About Me

Yogesh Haribhau Kulkarni

Bio:

- ▶ 20+ years in CAD/Engineering software development
- ▶ Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- ▶ Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- ▶ Feel free to follow me at:
 - ▶ Github (github.com/yogeshhk)
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KNN

K-Nearest Neighbors (KNN)

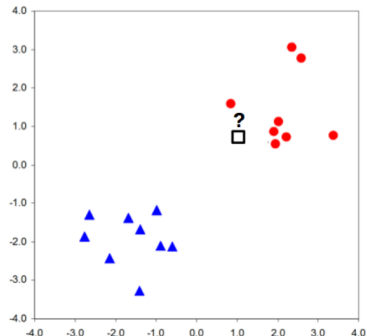
- ▶ Supervised Classification.
- ▶ Algorithm stores all available cases and classifies new cases by a majority vote of its k neighbors.

K-Nearest Neighbors (KNN)

- ▶ “A man is known by the company he keeps”
- ▶ KNN can easily be mapped to our real lives. If you want to learn about a person, of whom you have no information, you might like to find out about his close friends and the circles he moves in and gain access to his/her information!

Intuition for kNN

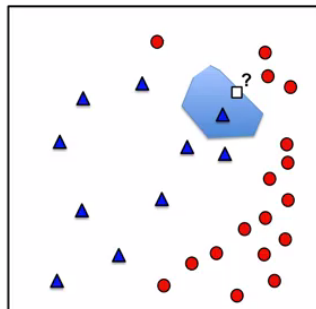
- ▶ Set of points (x,y) and two classes (red/blue)
- ▶ Is the box red or blue?
- ▶ How did you guess? By Decision Tree? or by SVM?
- ▶ You saw, nearby points are red.



Neighborhood is the basis of KNN.

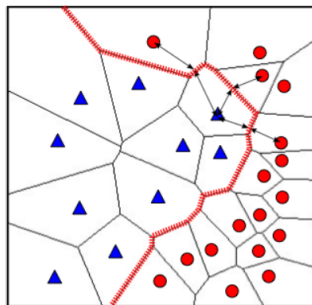
Nearest-neighbor classification

- ▶ Use the intuition to classify a new point x : find the most similar training example x' ; predict its class y'
- ▶ This is true for all the points shown in blue patch/cell.
- ▶ Every training example is going to have a small cell around it.
- ▶ All cells together are called as Voronoi Diagram.
- ▶ Examples: Post office region demarcation, Utility stations, etc.



Nearest-neighbor classification

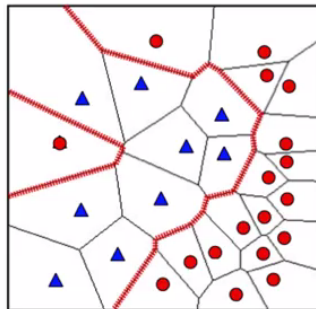
- ▶ Each cell is such that points in it are closer to its own Centroid than any other Centroids.
- ▶ Each cell corresponds to a training example.
- ▶ Classification boundary separates the cells based on classes
- ▶ See how a Non-linear complex partitioning is possible, than just linear or margin separation



Nearest-neighbor classification

Limitations

- ▶ But, if you have a Outlier, say the red shown, then whole classification goes for a toss.
- ▶ Boundary becomes too complicated.
- ▶ So, Sensitive to Outliers.
- ▶ Remedy: to look at more than one neighbors to classify yourself.
- ▶ Majority Voting. Adds stability.



Scaling Issues

- ▶ Attributes may have to be scaled to prevent distance measures from being dominated by one of the attributes
- ▶ Example, with three dimensions
 - ▶ height of a person may vary from 1.5m to 1.8m
 - ▶ weight of a person may vary from 90lb to 300lb
 - ▶ income of a person may vary from \$10K to \$1M
- ▶ Income will dominate if these variables aren't standardized.

Standardization

- ▶ Treat all features “equally” so that one feature doesn't dominate the others
- ▶ Common treatment to all variables
 - ▶ Standardize each variable: Mean = 0, Standard Deviation = 1
 - ▶ Normalize each variable: Max = 1, Min = 0

KNN Algorithm

Classification Algorithm

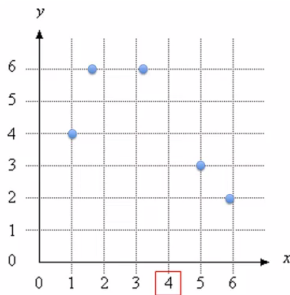
- ▶ Given: training examples (x_i, y_i) and test point x to classify.
- ▶ Here, there is no traditional training phase, where model is built.
- ▶ Compute distance $D(x, x_i)$ to every training example x_i
- ▶ Select k closest instances $x_{i1} \dots x_{ik}$ and their labels $y_{i1} \dots y_{ik}$
- ▶ Output the class y^* which is most frequent in $y_{i1} \dots y_{ik}$

KNN as Regression

- ▶ Given: training examples (x_i, y_i) with y_i as real valued, like profits, price, etc. and test point x to predict the target value.
- ▶ Here is no traditional training phase, where model is built.
- ▶ Compute distance $D(x, x_i)$ to every training example x_i
- ▶ Select k closest instances $x_{i1} \dots x_{ik}$ and their labels $y_{i1} \dots y_{ik}$
- ▶ Output the target y^* which is MEAN of $y_{i1} \dots y_{ik}$

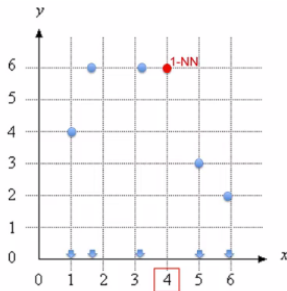
KNN Regression Example: 1d

- Given: Some set of points (x_i, y_i) . The function/relation between them is unknown.
- Goal: given, unseen x , find its y value.



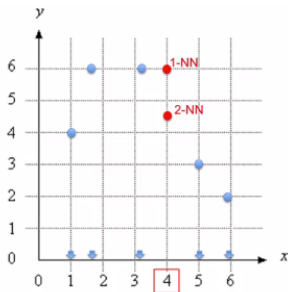
KNN Regression Example: 1d

- ▶ Who are the neighbours of $x = 4$ on x axis?
- ▶ If we want only one nearest neighbour (1-NN) then 3.2 is nearest, its y is 6, so that's the final prediction for $x = 4$.



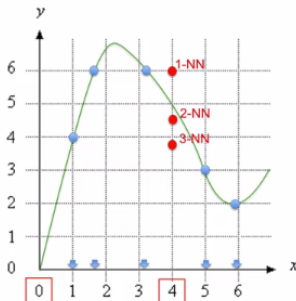
KNN Regression Example: 1d

- ▶ If we find 2-NN then 3.2 and 5 are the neighbours
- ▶ Their y s are 6 and 3.
- ▶ Average of those y s is 4.5. So for $x = 4$, y is predicted as 4.5



KNN Regression Example: 1d

- ▶ Its easy to see that if you go on adding neighbours, you may not be more accurate all the time.
- ▶ If you take all the neighbours?
- ▶ As it is just average of those many y values around.
- ▶ Is that good? No. There is one sweet spot (tradeoff).

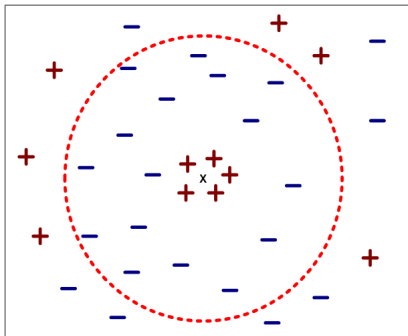


KNN Regression Example: 1d

- ▶ So, if you are in the middle, then its interpolation, works.
- ▶ If you are at boundaries then, its hard to predict. Extrapolation
- ▶ Find for $x = 0$ with 1-NN, 2NN.

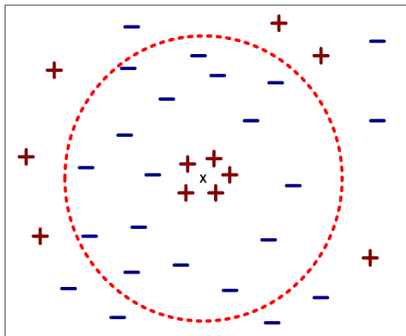
Choosing the right k

- Say, we are classifying middle point, into either +s or -s
- Depending on k (ie number of neighbours), it may get classified as + or -
- Say if k is upto 5, then it would be +, if say, 20 then it would be -ve



Choosing the right k

- ▶ If k is too small, sensitive to noise points in the training data
- ▶ Susceptible to overfitting
- ▶ If k is too large, neighborhood may include points from other classes
- ▶ Susceptible to misclassification



Choosing the right k

- ▶ Do, cross validation.
- ▶ Take validation set out of training set. Make it separate, not part of Training set any more.
- ▶ Try different k's on the validation set.
- ▶ You will predict y for every x in the test set.
- ▶ As you know the correct labels, see which k gives best result.

Similarity Criterion - Distance

Distance measures

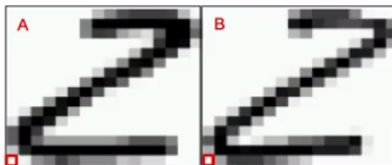
- ▶ Crucial component
- ▶ Finds similarity value between two data points.
- ▶ Types: Euclidean (for floats), Hamming (for enums), Minkowski (general form)

Distances

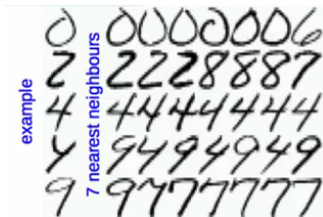
- ▶ The case being assigned to the class is most common among its K nearest neighbors measured by a distance function.
- ▶ These distance functions can be Euclidean, Manhattan, Minkowski and Hamming distance.

Example

- ▶ Handwritten digit recognition
- ▶ 16×16 bitmaps
- ▶ Euclidean Distance over pixels between testing bitmap A and training bitmap B. $D(A, B) = \sqrt{\sum_r \sum_c (A_{r,c} - B_{r,c})^2}$
- ▶ Find 7 nearest training examples, their known digit labels.
- ▶ Find most occurring among those labels



Example



- ▶ For “0”, max vote goes to “0”.
- ▶ For “2”, there is a tie, may add more neighbors to decide.
- ▶ For “4”, things may fail.
- ▶ Simple algorithm, but impressive results.
- ▶ Accuracy:
 - ▶ 7-NN: 95.2%
 - ▶ SVM: 95.8%
 - ▶ Humans: 97.5%, Humans are not perfect either!!

Distance measures (Recap)

- ▶ Euclidean is for numerical features: $\sqrt{\sum (x_d - x'_d)^2}$
- ▶ Hamming is for categorical features: $\sum 1_{x_d \neq x'_d}$ ie number of attributes, where x, x' differ
- ▶ Minkowski distance: $\sqrt[p]{\sum |x_d - x'_d|^p}$
 - ▶ $p = 2$: Euclidean
 - ▶ $p = 1$: Manhattan
 - ▶ $p = 0$: Hamming

INTUITION

Minkowski is just one dial (p) that stretches between two familiar rulers: at $p = 1$ you add up the “city block” steps along each axis (Manhattan), while at $p = 2$ you measure the straight-line distance (Euclidean). Larger differences get penalized more heavily as p grows, so the choice of p is really a choice about how harshly you want to punish a single large mismatch versus several small ones.

Implementing KNN

Writing our Own KNN from Scratch

A machine learning algorithm usually consists of 2 main steps:

- ▶ A training step that takes as input the training data X and the corresponding target y and outputs a learned model h
- ▶ A predict step that takes as input new and unseen observations and uses the function h to output their corresponding responses.

Writing our Own KNN from Scratch

In the case of KNN, which as discussed earlier, is a lazy algorithm, the training block reduces to just memorizing the training data

```
1 import numpy as np
  from collections import Counter
3
4 def train(X_train, y_train):
5     # do nothing
  return
```

Writing our Own KNN from Scratch

Now we need to write the predict method which must do the following:

- ▶ It needs to compute the euclidean distance between the “new” observation and all the data points in the training set.
- ▶ It must then select the K nearest ones and perform a majority vote.
- ▶ It then assigns the corresponding label to the observation.

Writing our Own KNN from Scratch

```
def predict(X_train, y_train, x_test, k):  
2     distances = []  
3     targets = []  
4  
5     for i in range(len(X_train)):  
6  
7         distance = np.sqrt(np.sum(np.square(x_test -  
8                                     X_train[i, :])))  
9         distances.append([distance, i])  
10  
11     distances = sorted(distances)  
12  
13     for i in range(k):  
14         index = distances[i][1]  
15         targets.append(y_train[index])  
16  
17     return Counter(targets).most_common(1)[0][0]
```

Writing our Own KNN from Scratch

Putting it all together, we can define the function `KNearestNeighbor`, which loops over every test example and makes a prediction.

```
1 def kNearestNeighbor(X_train,y_train,X_test, predictions, k):  
    train(X_train, y_train)  
3  
    for i in range(len(X_test)):  
5        predictions.append(predict(X_train, y_train,  
                                   X_test[i, :], k))
```

Writing our Own KNN from Scratch

Let's go ahead and run our algorithm

```
1 from sklearn.metrics import accuracy_score
2
3 # making our predictions
4 predictions = []
5
6 kNearestNeighbor(X_train, y_train, X_test, predictions, 7)
7
8 # transform the list into an array
9 predictions = np.asarray(predictions)
10
11 # evaluating accuracy
12 accuracy = accuracy_score(y_test, predictions)
13 print('\n\nThe accuracy is %d%%' % (accuracy * 100))
```

Optimizing KNN

- ▶ Comparing a query point a in d dimensions against n training examples computes with a runtime of $O(nd)$, which can cause lag as points reach millions or billions.
- ▶ Popular choices to speed up KNN include:
 - ▶ Voronoi Diagrams: partitioning plane into regions based on distance to points in a specific subset of the plane
 - ▶ Grid Indexes: carve up space into d -dimensional boxes or grids and calculate the NN in the same cell as the point
 - ▶ Locality Sensitive Hashing (LSH): abandons the idea of finding the exact nearest neighbors. Instead, batch up nearby points to quickly find the most appropriate bucket B for our query point.

INTUITION

All three tricks trade a bit of exactness for speed by pre-organizing the training points so that a query only needs to look at a small nearby subset instead of every single point. Voronoi/Grid indexes still find the true nearest neighbor but narrow the search space first; LSH goes a step further and accepts an approximate answer in exchange for even less work – a reasonable trade when “close enough” neighbors are good enough for the task.

Summary

Final Thoughts on Nearest Neighbors

- ▶ Nearest-neighbors classification is part of a more general technique called instance-based learning
- ▶ Use specific instances for prediction, rather than a model
- ▶ Nearest-neighbors is a lazy learner
- ▶ Performing the classification can be relatively computationally expensive
- ▶ No model is learned up-front

Final Thoughts on Nearest Neighbors

Pros

- ▶ The most attractive features of the K-nearest neighbor algorithm is that is simple to understand and easy to implement.
- ▶ With zero to little training time, it can be a useful tool for off-the-bat analysis of some data set you are planning to run more complex algorithms on.
- ▶ Furthermore, KNN works just as easily with multiclass data sets whereas other algorithms are hardcoded for the binary setting.
- ▶ Finally, as we mentioned earlier, the non-parametric nature of KNN gives it an edge in certain settings where the data may be highly “unusual”.

Final Thoughts on Nearest Neighbors

Cons

- ▶ Computationally expensive testing phase which is impractical in industry settings.
- ▶ Can suffer from skewed class distributions. For example, if a certain class is very frequent in the training set, it will tend to dominate the majority voting of the new example (large number = more common).
- ▶ Finally, the accuracy of KNN can be severely degraded with high-dimension data because there is little difference between the nearest and farthest neighbor.

K-Nearest Neighbors (Recap)

- ▶ k = parameter, chosen by analyst
- ▶ For a given test instance, use the k “closest” points (nearest neighbors) for performing classification
- ▶ “closest” points: defined by some proximity metric, such as Euclidean Distance
- ▶ Requires three things
 - ▶ The set of stored records
 - ▶ Distance Metric to compute distance between records
 - ▶ The value of k , the number of nearest neighbors to retrieve

Quick Check: K-Nearest Neighbor

THINK ABOUT IT

Suppose you train a KNN model to predict house prices using square footage (values in the thousands) and number of bedrooms (typically 1-5), without scaling either feature. Which feature will dominate the distance calculation, and why does this make KNN especially sensitive to feature scaling compared to a Decision Tree trained on the same raw data?

ANSWER

Square footage will dominate, since distances are computed on raw magnitudes and a difference of a few hundred square feet dwarfs a difference of one bedroom. KNN is sensitive to this because every feature contributes directly to a single geometric distance number; a Decision Tree instead considers one feature at a time when choosing splits, so an unscaled feature doesn't drown out the others the way it does for a distance metric.

Quick Check: K-Nearest Neighbor

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K-Nearest Neighbors with Scikit-Learn

Knn Template Code

```
1 from sklearn.neighbors import KNeighborsClassifier
3 #Assumed you have, X (predictor) and Y (target)
  # for training and x_test(predictor) of test
5
7 model = KNeighborsClassifier(n_neighbors=6)
9
10 model.fit(X, y)
11
12 predicted= model.predict(x_test)
```

K-Nearest Neighbors

(Ref: Machine Learning Algorithm Recipes in scikit-learn - Jason Brownlee)

```
1 # k-Nearest Neighbor
2 from sklearn import datasets
3 from sklearn import metrics
4 from sklearn.neighbors import KNeighborsClassifier
5 # load iris the datasets
6 dataset = datasets.load_iris()
7 # fit a k-nearest neighbor model to the data
8 model = KNeighborsClassifier()
9 model.fit(dataset.data, dataset.target)
10 print(model)
11 # make predictions
12 expected = dataset.target
13 predicted = model.predict(dataset.data)
14 # summarize the fit of the model
15 print(metrics.classification_report(expected, predicted))
16 print(metrics.confusion_matrix(expected, predicted))
```


Test case: Iris Flowers

Review of the iris dataset

```
import pandas as pd
2
url = 'https://raw.githubusercontent.com/jbrownlee/Datasets/master/iris.csv'
4
col_names = ['sepal_length', 'sepal_width', 'petal_length',
6             'petal_width', 'species']
8
iris = pd.read_csv(url, header=None, names=col_names)
```

Review of the iris dataset

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

- ▶ 150 observations (n=150): each observation is one iris flower
- ▶ 4 features: sepal length, sepal width, petal length, petal width
- ▶ Response: iris species
- ▶ Classification problem since response is categorical

Encoding Iris dataset

Encoding of the Target

```
1 iris['species_num'] = iris.species.map({'Iris-setosa':0,  
3                                     'Iris-versicolor':1,  
                                     'Iris-virginica':2})
```

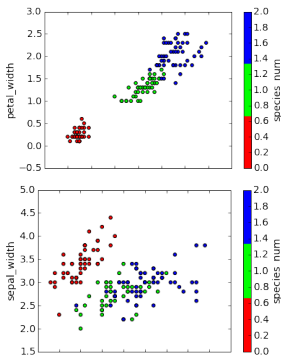
Plot Iris dataset

Exploratory Data Analysis

```
1 # create a scatter plot of PETAL LENGTH versus PETAL WIDTH  
  # and color by SPECIES  
3 iris.plot(kind='scatter', x='petal_length', y='petal_width', c='species_num',  
           colormap=cmap_bold)  
  
5 # create a scatter plot of SEPAL LENGTH versus SEPAL WIDTH  
  # and color by SPECIES  
7 iris.plot(kind='scatter', x='sepal_length', y='sepal_width', c='species_num',  
           colormap=cmap_bold)
```

Plot Iris dataset

Exploratory Data Analysis



- ▶ setosas have small petals, versicolor have medium sized petals and virginica have the largest petals.
- ▶ Furthermore, setosas seem to have shorter and wider sepals than the other two classes.

KNN modeling

```
1 feature_cols = ['sepal_length', 'sepal_width',  
                 'petal_length', 'petal_width']  
3  
4 X = iris[feature_cols]  
5 y = iris.species_num  
6  
7 from sklearn.neighbors import KNeighborsClassifier  
8  
9 knn = KNeighborsClassifier(n_neighbors=1)  
10  
11 knn.fit(X, y)  
12  
13 knn.predict([[3, 5, 4, 2]]) # 2  
14  
15 X_new = [[3, 5, 4, 2], [5, 4, 3, 2]]  
    knn.predict(X_new) # 2,1
```

Measuring Error

With $k = 1$

```
1 from sklearn.model_selection import train_test_split
2
3 X_train, X_test, Y_train, Y_test = train_test_split(
4     X,y, test_size=0.4, random_state=4)
5
6 actual = Y_test
7
8 knn = KNeighborsClassifier(n_neighbors=1)
9 knn.fit(X_train, Y_train)
10 expected = knn.predict(X_test) #predictions
11
12 from sklearn import metrics
13 score_1 = metrics.accuracy_score(actual, expected)
14 print(score_1) #0.94
```


Measuring Error

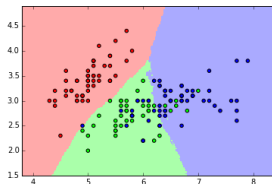
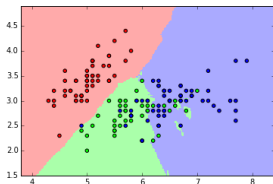
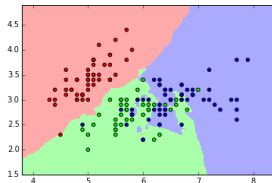
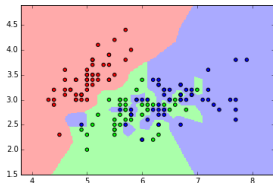
Try range of k values

```
1 import matplotlib.pyplot as plt
3 scores = []
  k_range = range(1,26)
5 for k in k_range:
    knn=KNeighborsClassifier(n_neighbors=k)
7    knn.fit(X_train, Y_train)
    actual = knn.predict(X_test)
9    scores.append(metrics.accuracy_score(Y_test, actual))

11 plt.xlabel('Value of K for KNN')
   plt.ylabel('Testing Accuracy')
13 plt.plot(k_range, scores)
   plt.show()
```

Effect of K

K=1,5, 15,50



Test case: NBA Players

Exercise

A look at the data, Each row in the data contains information on how a player performed in the 2013-2014 NBA season.

```
import pandas
2 with open("nba_2013.csv", 'r') as csvfile:
    nba = pandas.read_csv(csvfile)
4
# The names of all the columns in the data.
6 print(nba.columns.values)
```

Exercise

Here are some selected columns from the data:

- ▶ player - name of the player
- ▶ pos - the position of the player
- ▶ g - number of games the player was in
- ▶ gs - number of games the player started
- ▶ pts - total points the player scored

There are many more columns in the data, mostly containing information about average player game performance over the course of the season.

Exercise

Study distance: find the most similar NBA players to LeBron James.

```
import math

2
# Extract LeBron's row to use as the reference point
4 selected_player = nba[nba["player"] == "LeBron James"].iloc[0]

6 # Numeric columns to measure player similarity across stats
distance_columns = ['age', 'g', 'gs', 'mp', 'fg', 'fga',
8                    'fg.', 'x3p', 'x3pa', 'x3p.', 'x2p',
                    'x2pa', 'x2p.', 'efg.', 'ft', 'fta',
10                   'ft.', 'orb', 'drb', 'trb', 'ast', 'stl',
                    'blk', 'tov', 'pf', 'pts']

12
def euclidean_distance(row):
14     # Sum squared differences across all stat columns vs LeBron
    inner_value = 0
16     for k in distance_columns:
        inner_value += (row[k] - selected_player[k]) ** 2
18     return math.sqrt(inner_value)

20 # Compute distance from every player to LeBron row-by-row
lebron_distance = nba.apply(euclidean_distance, axis=1)
```

Exercise

Normalizing columns

```
1 # Select only the numeric columns from the NBA dataset
  nba_numeric = nba[distance_columns]
3
  # Normalize all of the numeric columns
5 nba_normalized = (nba_numeric - nba_numeric.mean()) / nba_numeric.std()
7
  # Fill in NA values in nba_normalized
9 nba_normalized.fillna(0, inplace=True)
```

Exercise

Use the `distance.euclidean` function from `scipy.spatial`, a much faster way to calculate euclidean distance.

```
1 from scipy.spatial import distance
3 # Find the normalized vector for lebron james.
  lebron_normalized = nba_normalized[nba["player"] == "LeBron James"]
5
  # Find the distance between lebron james and everyone else.
7 euclidean_distances = nba_normalized.apply(lambda row: distance.euclidean(row,
  lebron_normalized), axis=1)
9
  # Create a new dataframe with distances.
  distance_frame = pandas.DataFrame(data={"dist": euclidean_distances, "idx":
  euclidean_distances.index})
11 distance_frame.sort_values("dist", inplace=True)
  # Find the most similar player to lebron (the lowest distance to lebron is
  lebron, the second smallest is the most similar non-lebron player)
13 second_smallest = distance_frame.iloc[1]["idx"]
  most_similar_to_lebron = nba.loc[int(second_smallest)]["player"]
```


Exercise

Generating training and testing sets

```
import random
2 from numpy.random import permutation

4 # Randomly shuffle the index of nba.
  random_indices = permutation(nba.index)

6 # Set a cutoff for how many items we want in the test set (in this case 1/3 of
  the items)

8 test_cutoff = math.floor(len(nba)/3)

10 # Generate the test set by taking the first 1/3 of the randomly shuffled
  indices.
  test = nba.loc[random_indices[1:test_cutoff]]

12 # Generate the train set with the rest of the data.

14 train = nba.loc[random_indices[test_cutoff:]]
```

Exercise

Using sklearn for k nearest neighbors

```
1 x_columns = ['age', 'g', 'gs', 'mp', 'fg', 'fga', 'fg.', 'x3p', 'x3pa',  
2             'x3p.', 'x2p', 'x2pa', 'x2p.', 'efg.', 'ft', 'fta', 'ft.', 'orb', 'drb',  
3             'trb', 'ast', 'stl', 'blk', 'tov', 'pf']  
4  
5 y_column = ["pts"]  
6  
7 from sklearn.neighbors import KNeighborsRegressor  
8  
9 knn = KNeighborsRegressor(n_neighbors=5)  
10  
11 knn.fit(train[x_columns], train[y_column])  
12  
13 predictions = knn.predict(test[x_columns])
```

Exercise

Computing error

```
1 actual = test[y_column]
3 mse = (((predictions - actual) ** 2).sum()) / len(predictions)
```

Classifier Comparison

Eager Learners

- ▶ Decision Trees, SVMs
- ▶ Model Building: potentially slow
- ▶ Classifying Test Instance: fast
- ▶ finding a global model that fits the entire input space

Lazy Learners

- ▶ Nearest Neighbors
- ▶ Model Building: fast (because there is none!)
- ▶ Classifying Test Instance: slow
- ▶ classification decisions are made locally (small k values), and are more susceptible to noise

Quick Check: KNN with Scikit-Learn

THINK ABOUT IT

If you evaluate a $k = 1$ KNN model by measuring its accuracy on the very same data it was fit on, it will score close to 100%. Why is this measurement almost meaningless, and how does the “Measuring Error” code above (using `train_test_split`) fix it?

ANSWER

With $k = 1$, every training point's single nearest neighbor is itself, so it trivially predicts its own label – the score reflects memorization, not generalization to unseen data. Splitting into train/test sets forces the model to predict labels for points it never saw during fitting, which is the only way to estimate how it will perform on new data.

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Thanks ...

- ▶ Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
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